

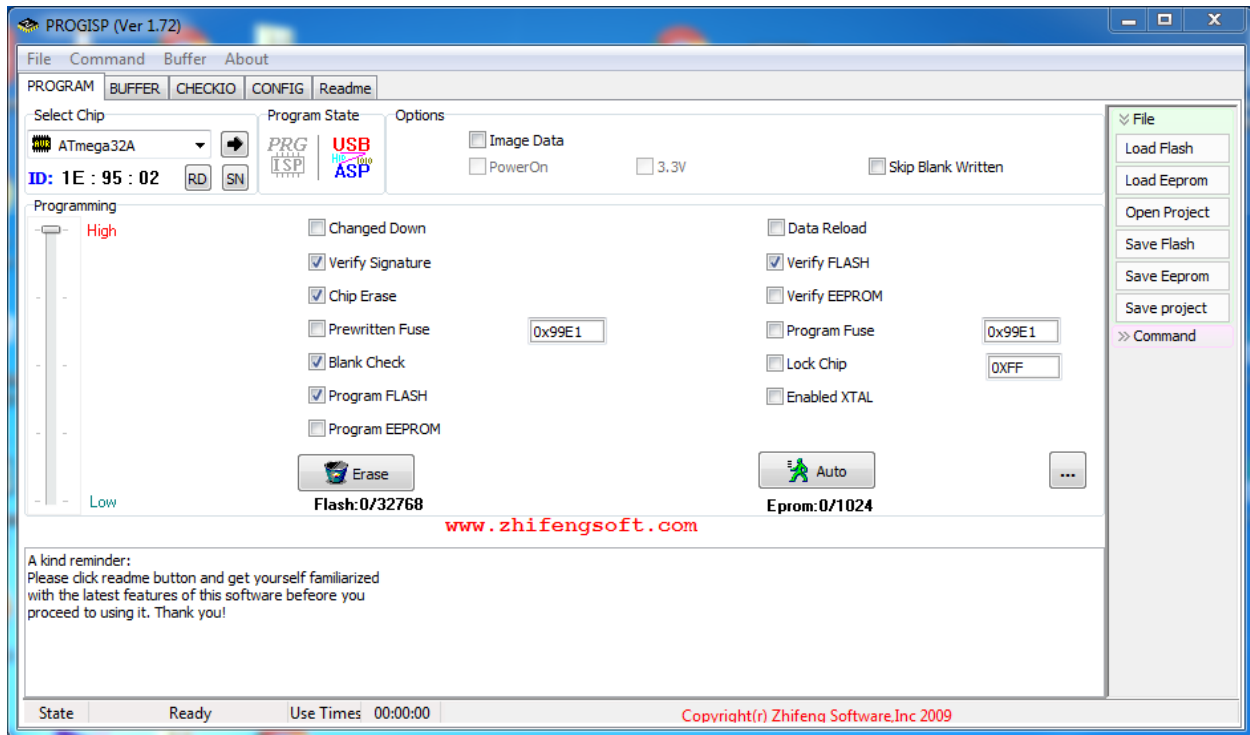
Practical Hardware Assignment #1

Familiarization with AVR microcontrollers

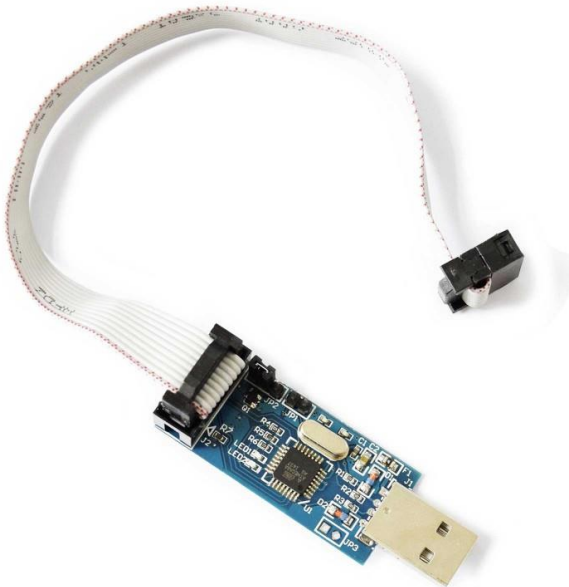
(due date during lab in week 4)

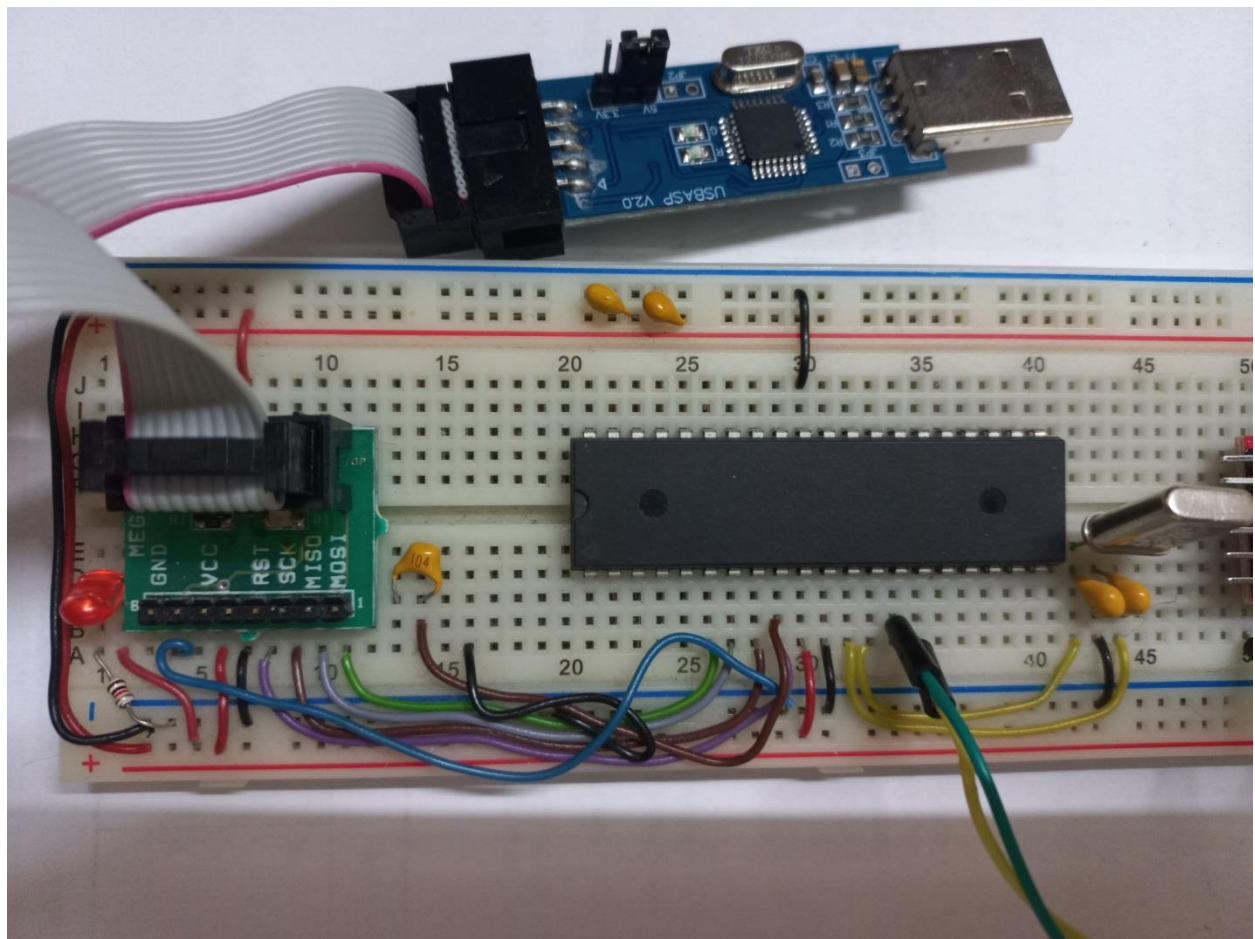
- 1- Install **Codevision** AVR C compiler
Version 3.14 will be provided to students
- 2- Get AVR ATMEGA16A/32A microcontroller and 11.0592MHz or 14.7456MHz or 16MHz crystal and connect the minimum system on a bread board.
- 3- Get **USBASP AVR Programmer** and install its driver and download the free programming software **Progisp1.72**
- 4- Write a simple C code that flashes a LED (1 sec ON and 1 sec OFF) on PORTA.0
- 5- Simulate The code using **Proteus**
- 6- Download the generated hex code to the microcontroller.

To program the micro controller we will use ProgISP (Ver 1.72) software



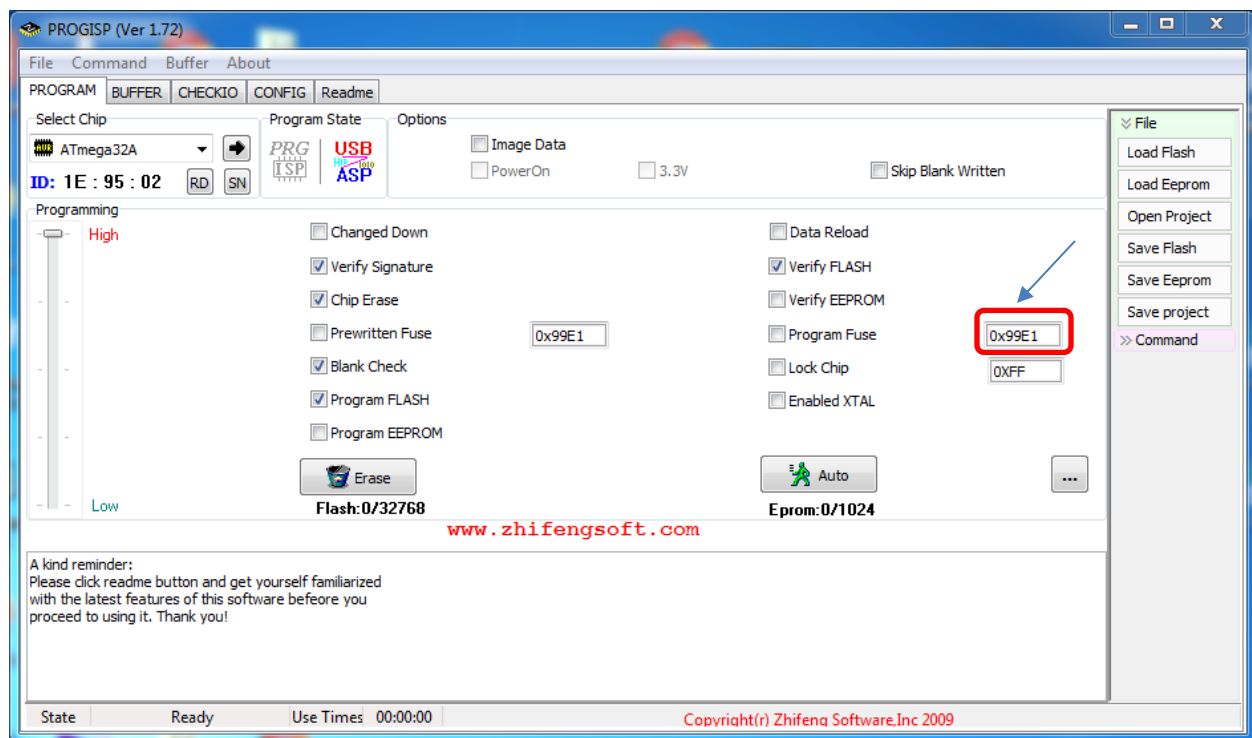
The programmer is the USBASP AVR programmer







Run the software and click on the text box where the arrow points



For a brand new controller

Notice the following fuses : CKSEL3 CKSEL2 CKSEL1 CKSEL0 having the value 0 0 0 1

Fuse&Lock
⌕

Low Fuse Bits	High Fuse Bits	Extend Fuse Bits	Lock Bits	Calibration
☒ BODLEVEL	☒ OCDEN	☐ NC	☒ NC	1.0 MHz 00
☒ BODEN	☐ JTAGEN	☐ NC	☒ NC	2.0 MHz 00
☒ SUT1	☐ SPIEN	☐ NC	☒ BLB12	4.0 MHz 00
☐ SUT0	☒ CKOPT	☐ NC	☒ BLB11	8.0 MHz 00
☐ CKSEL3	☒ EESAVE	☐ NC	☒ BLB02	
☐ CKSEL2	☐ BOOTSZ1	☐ NC	☒ BLB01	
☐ CKSEL1	☐ BOOTSZ0	☐ NC	☒ LB2	
☒ CKSEL0	☒ BOOTRST	☐ NC	☒ LB1	

ConfigBit

Navigation

LowValue

HighValue

ExtValue

Lock Value

They are responsible for choosing the type of clock used which is a Calibrated Internal RC Oscillator with a value of 1 MHz as seen from the following table

Table 9-8. Internal Calibrated RC Oscillator Operating Modes

CKSEL3:0	Nominal Frequency (MHz)
0001 ⁽¹⁾	1.0
0010	2.0
0011	4.0
0100	8.0

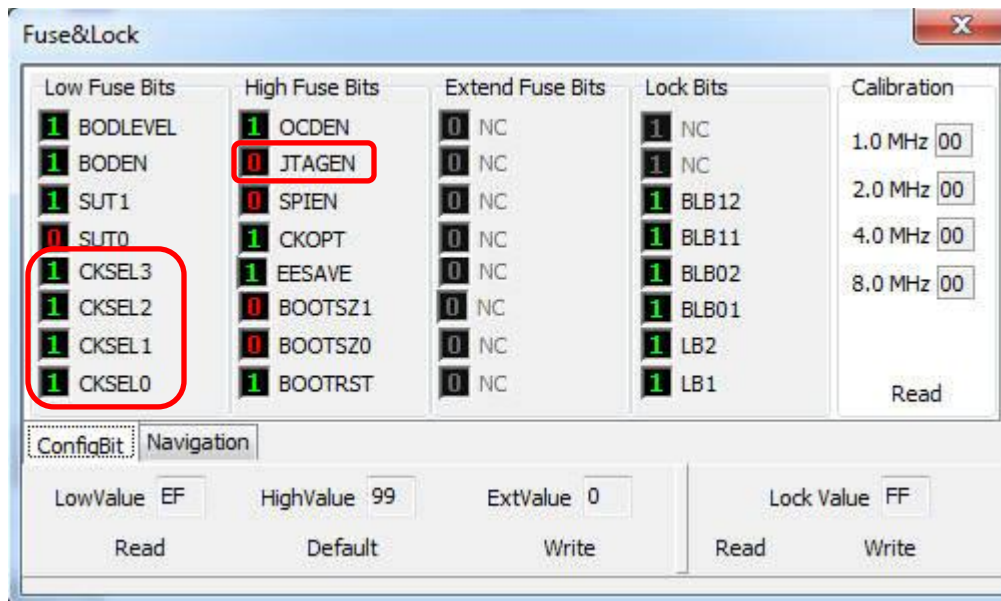
Note: 1. The device is shipped with this option selected.

The following table gives the different available options for the clock

Table 9-1. Device Clocking Options Select⁽¹⁾

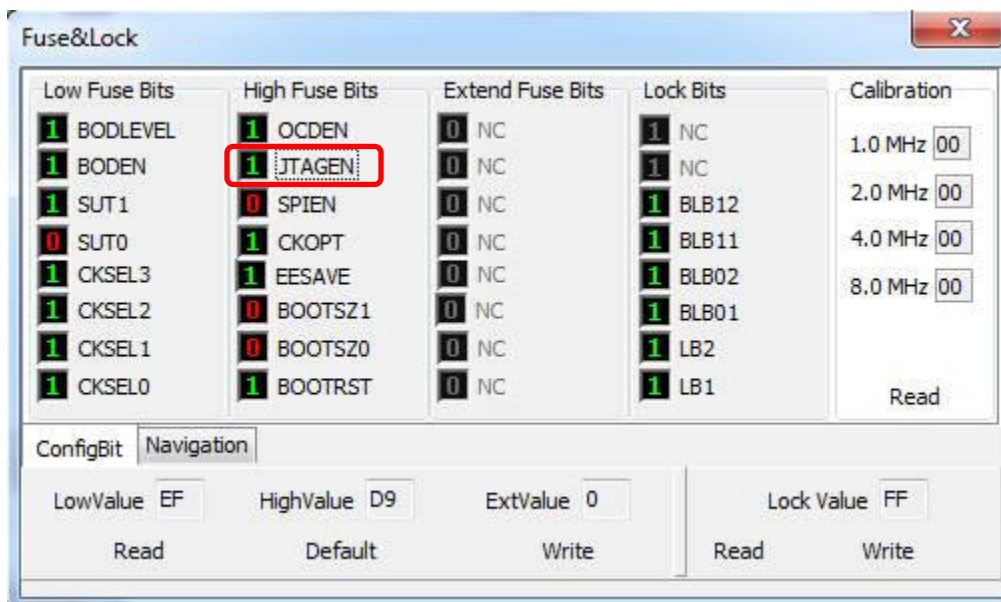
Device Clocking Option	CKSEL3:0
External Crystal/Ceramic Resonator	1111 - 1010
External Low-frequency Crystal	1001
External RC Oscillator	1000 - 0101
Calibrated Internal RC Oscillator	0100 - 0001
External Clock	0000

Since we want to use an external crystal oscillator above 8 MHz, we have to program the 4 bits of CKSEL to 1 1 1 1 as in the picture



Also notice the following fuse JTAGEN having the value 0 which means the JTAG is enabled so 4 bits of PORTC won't be available for use since they are used by the JTAG

So to be able to use all pins of port C we have to disable it by writing 1



Don't try to experiment with fuses at all unless you know what you are doing exactly as you might render the micro controller unusable